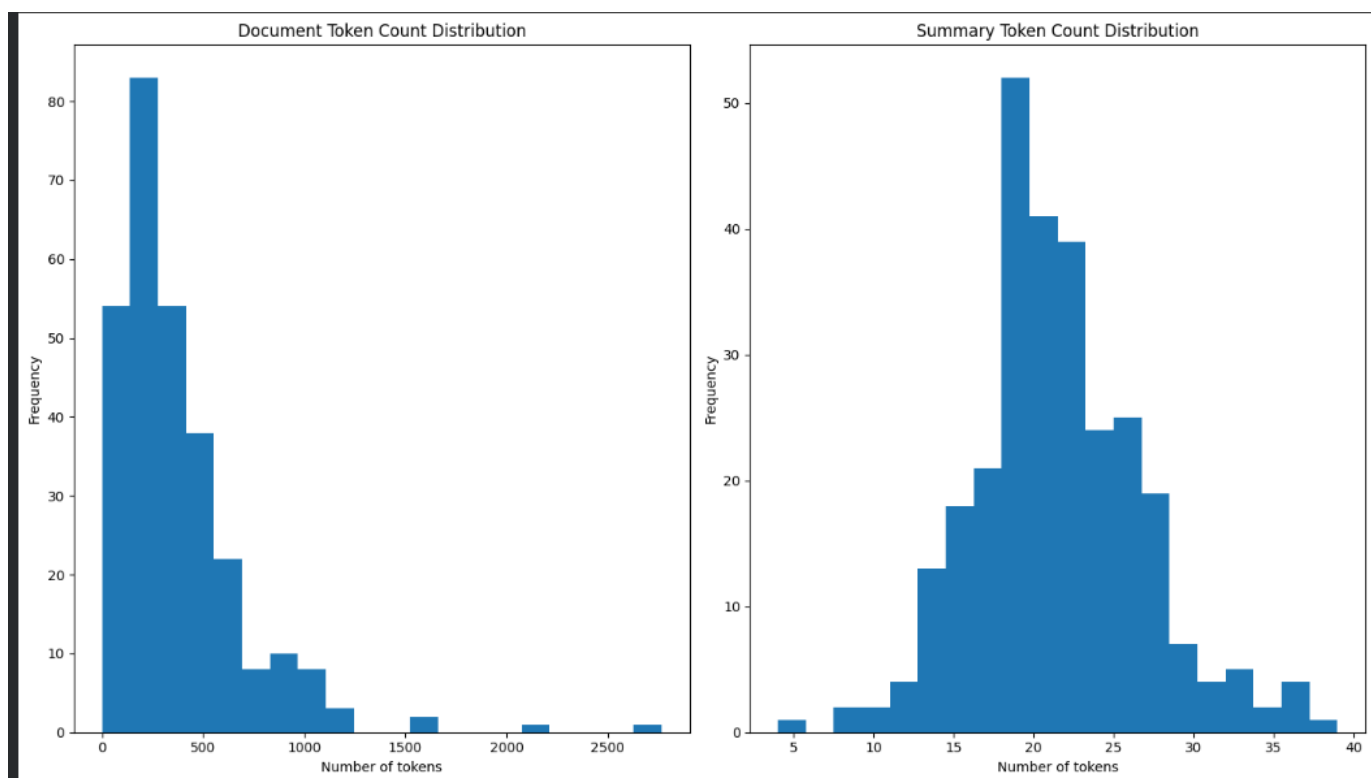




T5_Summarisation_and_Data_Generation

Question 1

Analyse the XSum summarisation dataset (Task 1). In your report, include the mean count and standard deviation for input and target documents for the indicated dataset. Plot histograms of the input document and target summary token counts. Comment on the lengths of input and target texts.



```
Dataset split: train
=====

Source Documents:
Mean token count      : 370.84
Standard deviation    : 299.88

Target Summaries:
Mean token count      : 21.00
Standard deviation    : 5.26

=====
Dataset split: validation
=====

Source Documents:
Mean token count      : 377.52
Standard deviation    : 325.83

Target Summaries:
Mean token count      : 21.33
Standard deviation    : 5.27
```

These results show that the input documents are quite longer than the target summaries and these documents are of different lengths which can be clearly seen by large standard deviation and token count value of document histogram meanwhile target summaries are short and consize with low standard deviation which tells they are almost around the same length . On average documents contain around 370 to 380 tokens while summaries are around 21 tokens. The summarization task requires compressing a large amount of information into a very short sentence usually called extreme summarization.

Question 2

Fine-tune and evaluate T5 for the XSum summarisation task (Task 2). In your report, include the final loss and validation ROUGE after five epochs of fine-tuning. Also, report outputs of the fine-tuned and not fine-tuned T5 models for the sample input document, as well as their ROUGE scores. Comment on the efficiency of fine-tuning taking those results into consideration.

```
Epoch 1: Training loss: 4.0661187171936035
Epoch 2: Training loss: 2.685080051422119
Epoch 3: Training loss: 3.1022140979766846
Epoch 4: Training loss: 2.7275619506835938
Epoch 5: Training loss: 2.887080669403076
```

Validation ROUGE:

```
{'rouge1': np.float64(0.25013430383046664), 'rouge2': np.float64(0.054476583246278375), 'rougeL': np.float64(0.18753827511721322), 'rougeLsum': np.float64(0.18793947006518857)}
```

Reference summary

A 90-minute match but only 47 minutes and 40 seconds of actual football.

Fine-tuned summary

The Premier League has been a fixture in the Premier League since the start of the season.

```
79]
✓ 0s metric.compute(predictions=[out_ft], references=[summary])

▼ {'rouge1': np.float64(0.12903225806451615),
   'rouge2': np.float64(0.0),
   'rougeL': np.float64(0.12903225806451615),
   'rougeLsum': np.float64(0.12903225806451615)}
```

Non-fine-tuned summary

a number of premier league games have seen less action this season. the league has been dominated by possession statistics. the league has been dominated by possession statistics

```
metric.compute(predictions=[out_zero_shot], references=[summary])

{'rouge1': np.float64(0.09523809523809523),
 'rouge2': np.float64(0.0),
 'rougeL': np.float64(0.09523809523809523),
 'rougeLsum': np.float64(0.09523809523809523)}
```

The performance of fine tuned model was compared with non fine tuned T5 model, the results are displayed above which shows that the fine tuned model achieved slightly higher score then non fine tuned .Fine tuing improved the models ability to generate summaries.However the improvement is relatively small and the generated summaries are still not very close to the reference summary.This may be because of limited training data or short training duration.

Question 3

Build and evaluate a T5-based model for data generation using keywords (Task 3). In your report, include the final loss after five epochs of training. Also, report the output of your model for the keywords provided in the script. Comment on the efficiency of the data generation procedure..

Final loss after 5 epochs of training

Training loss: 2.2884602546691895

Output of the model:

Hawick has been stranded in a hammer in Lamington via a commercial thoroughfare multi-agency network neglected and neglected by Hawick.

The data generation procedure was partially effective. The model was able to produce an output conditioned on the provided keywords which shows that it has learned some mapping from keywords to summary like text but the generated sentence is not fully coherent as it contains repetition and unnatural phrasing. So, the generation quality is limited.